

CEN 302 SOFTWARE ENGINEERING

BINF 3A

Group: 5

Phase III: Software Design and Modeling

1. Software Architecture

System Architecture

Describe the overall structure of your system. Explain how different parts of the application work together and what the system does.

TESS follows a three-tier architecture consisting of frontend, backend and database.

The frontend is the part of the system that students, employers and administrators interact with. It includes pages such as login/register, student dashboard, employer dashboard, job listings, application status, messaging and interview pages.

The backend provides REST APIs. It receives requests from the frontend, applies business logic, validates data, checks user roles and communicates with the database.

The database is PostgreSQL and stores all persistent system data, including users, students, employers, jobs, applications, availability schedules, messages, interviews and notifications.

General System Flow

- A student, employer, or admin interacts with the frontend.
- The frontend sends a request to the backend through REST APIs.
- The backend validates the request and checks the user's role and permissions.
- The backend communicates with the PostgreSQL database to retrieve or store required data.
- The database returns the requested data to the backend.
- The backend processes the result and sends a response back to the frontend.
- The frontend displays the result to the user.

Example Flow: Student Searching and Applying for a Job

- Student opens the job search page in the frontend.
- The frontend sends a request to the backend to retrieve suitable jobs.
- The backend checks the student's availability schedule from the database.
- The backend compares the student's availability with job working hours and requirements.
- If matching jobs exist, the backend sends the matching job list back to the frontend.
- The frontend displays the matching jobs to the student.
- The student selects a job and clicks Apply.
- The frontend sends the application request to the backend.
- The backend checks the selected job and student information in the database.
- The backend stores the application with status "Pending".
- The backend sends a confirmation response to the frontend.
- The frontend shows a success message to the student.

Example Flow: Employer Reviewing Applications

- Employer opens the employer dashboard.
- The frontend requests the list of applications for the employer's posted jobs.
- The backend checks the employer's jobs in the database.
- The backend retrieves applications linked to those jobs.
- The backend sends the applications back to the frontend.
- The employer reviews the student applications.
- If the employer accepts a candidate, the backend updates the application status and can generate interview details.
- The student is notified about the update.

Main Components

- Frontend: Displays the interface and allows users to interact with the system.
- Backend: Handles business logic, validation, REST APIs and system operations.
- Database: Stores and manages all system data.

1. Component Diagram (UML)

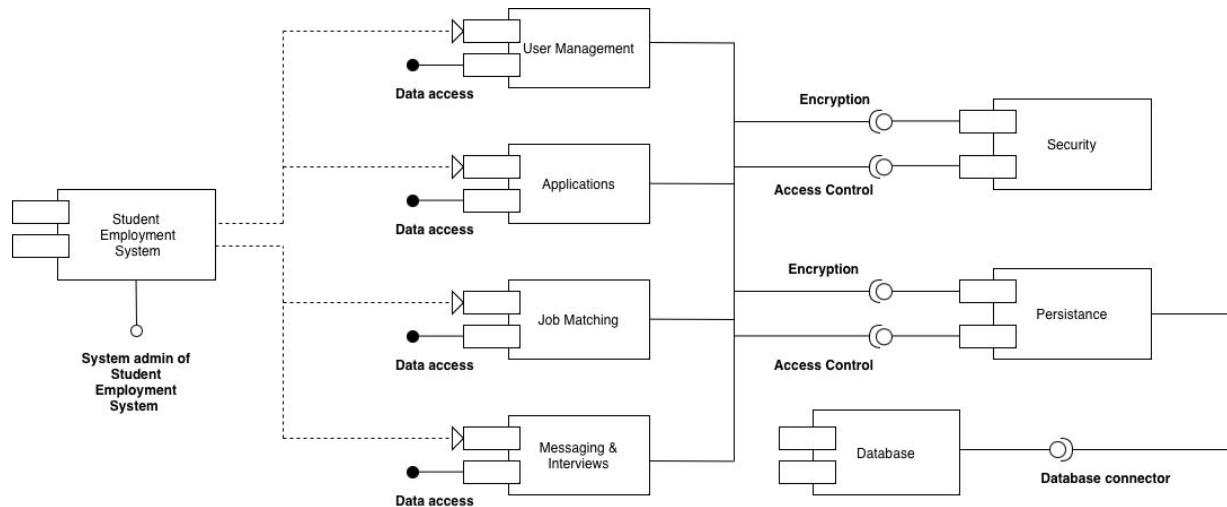


Figure 1: Component Diagram

User Management: manages user accounts, profiles, and authentication.

Applications: handles student job applications and submissions.

Job Matching: matches students with suitable job opportunities based in requirements.

Messaging & Interviews: manages text-based communication and real video communication.

Security: handles encryption and access control for system protection.

Persistence: manages storing and retrieving data reliably.

Database: stores all system data (users, jobs, applications and more).

Data Access: reads and writes data between components and the database.

Diagram Link: https://lucid.app/lucidchart/6b772a79-0d96-4c10-9f30-6e47c4e6642b/edit?viewport_loc=2524%2C3062%2C1918%2C1171%2C0_0&invitationId=inv_56a2d1a5-17ac-45a9-809e-2f07b1152ca1

2. Detailed Design

1. UML Class Diagram

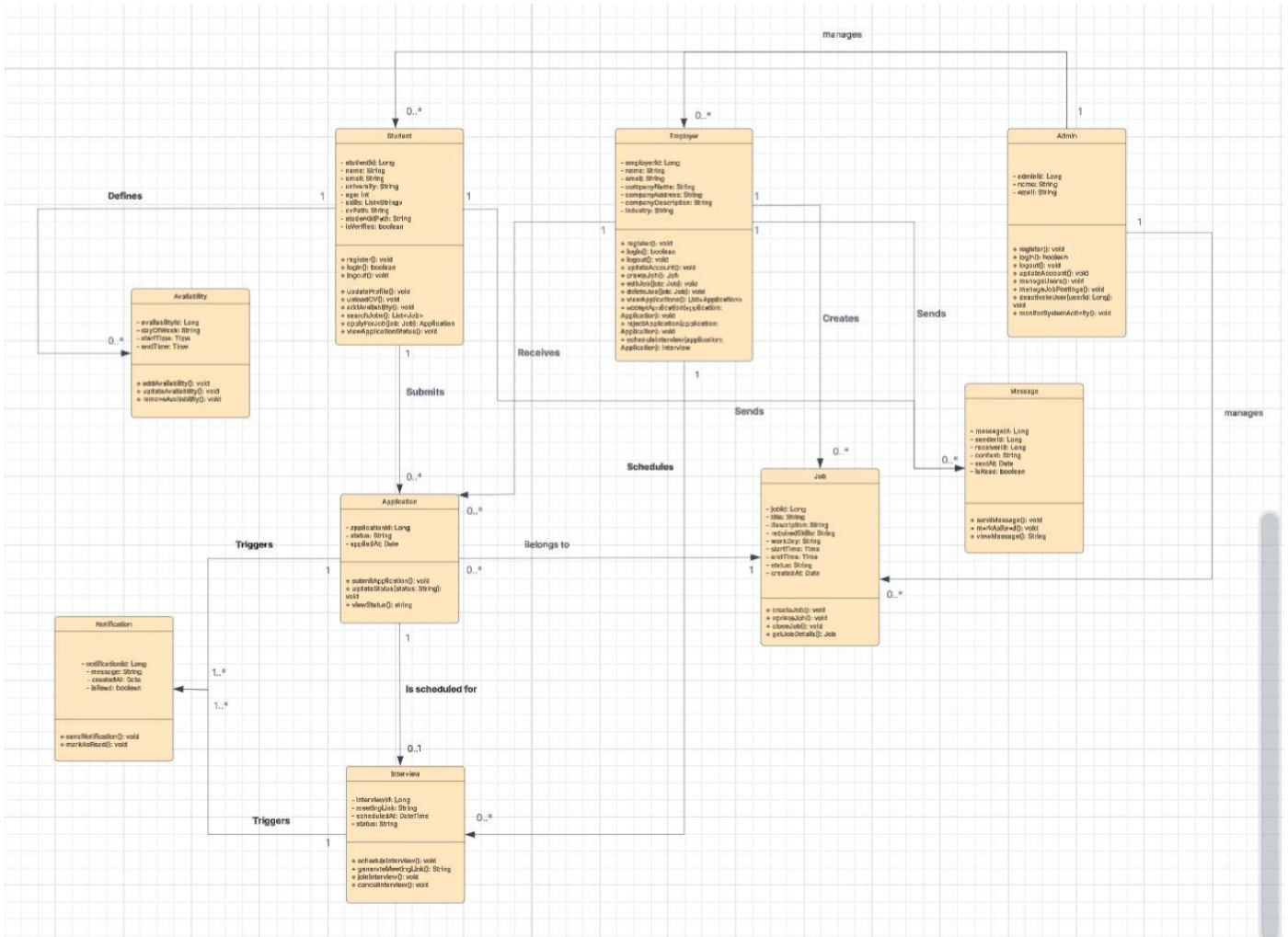


Figure 2: UML Class Diagram

Diagram Link: https://lucid.app/lucidchart/6b772a79-0d96-4c10-9f30-6e47c4e6642b/edit?viewport_loc=2524%2C3062%2C1918%2C1171%2C0_0&invitationId=inv_56a2d1a5-17ac-45a9-809e-2f07b1152ca1

2. Sequence diagrams

a. Sequence diagram (Students)

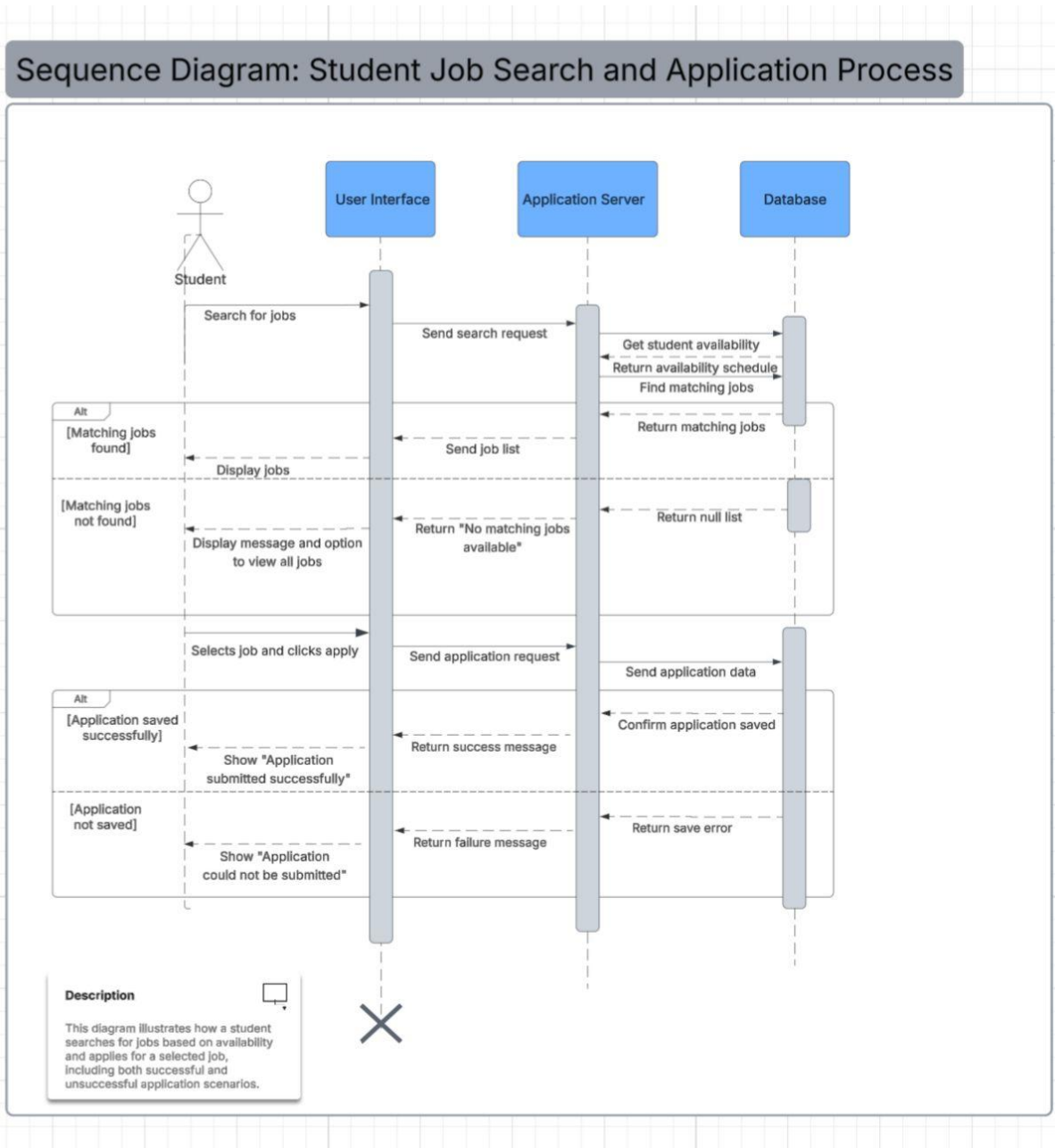


Figure 3: Sequence diagram (Students)

Diagram Link: https://lucid.app/lucidchart/6b772a79-0d96-4c10-9f30-6e47c4e6642b/edit?viewport_loc=2524%2C3062%2C1918%2C1171%2C0_0&invitationId=inv_56a2d1a5-17ac-45a9-809e-2f07b1152ca1

b. Sequence diagram (Employers)

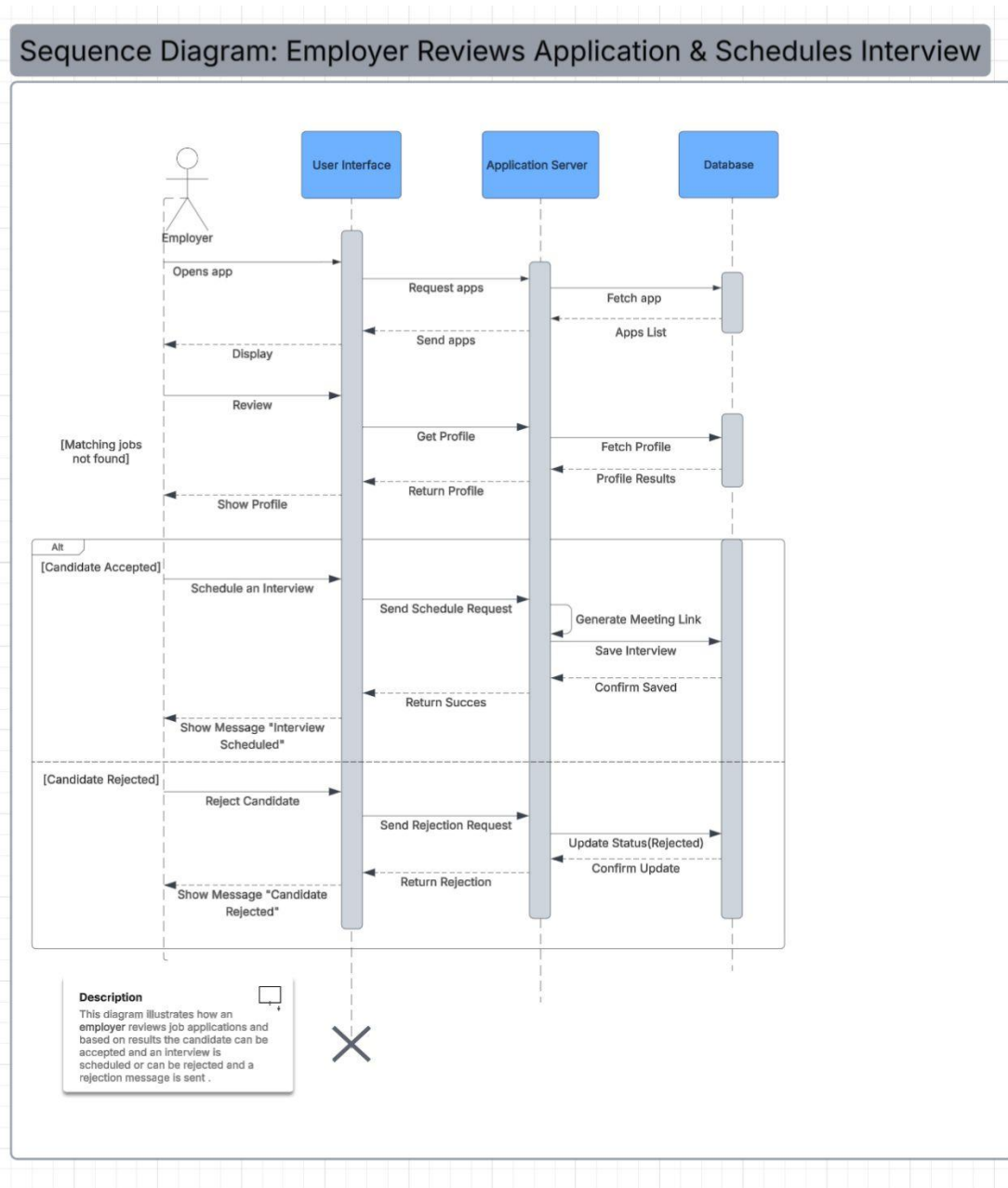


Figure 4: Sequence diagram (Employers)

Diagram Link: https://lucid.app/lucidchart/6b772a79-0d96-4c10-9f30-6e47c4e6642b/edit?viewport_loc=2524%2C3062%2C1918%2C1171%2C0_0&invitationId=inv_56a2d1a5-17ac-45a9-809e-2f07b1152ca1

3. Use Case Diagram

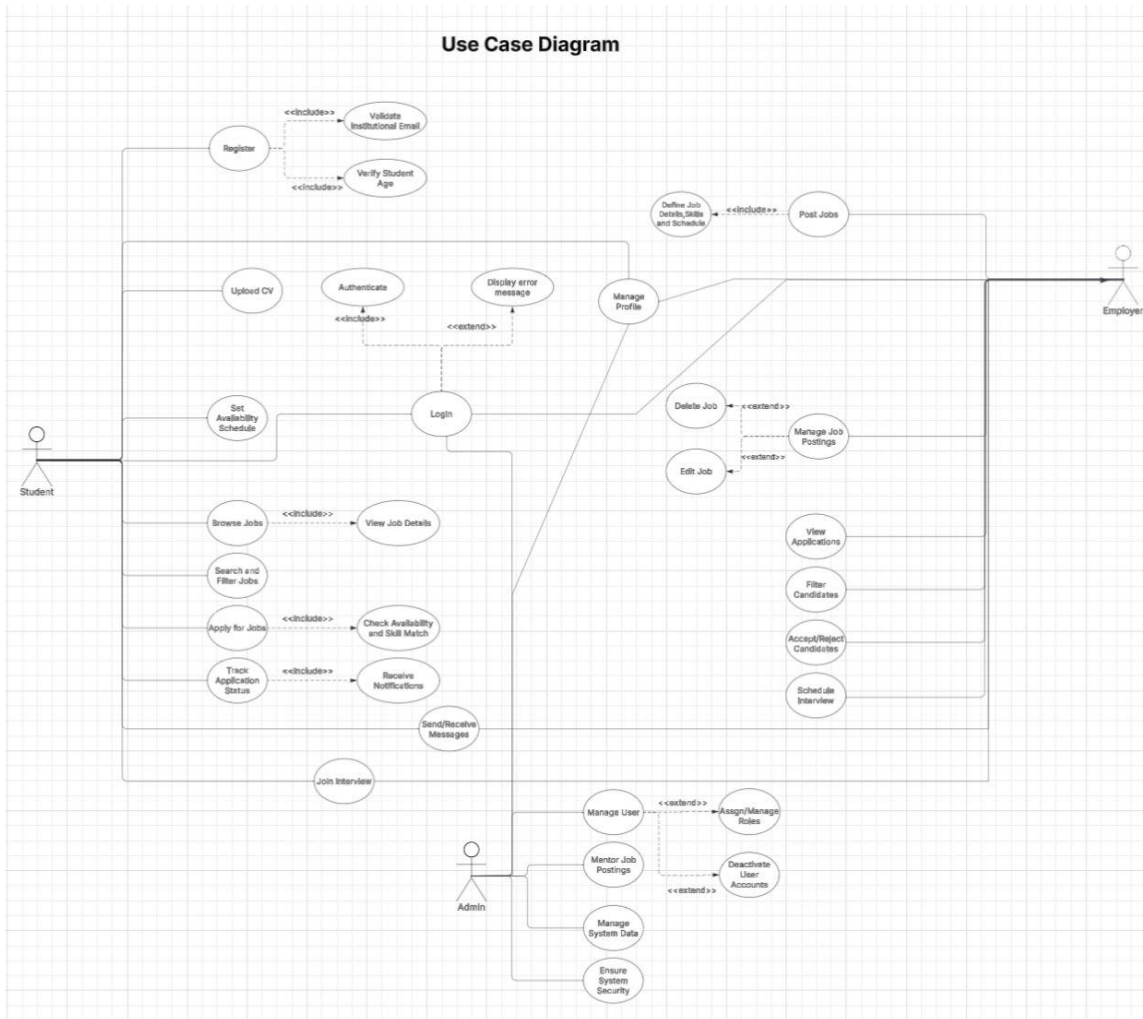


Figure 5: Use Case Diagram

Diagram Link: https://lucid.app/lucidchart/6b772a79-0d96-4c10-9f30-6e47c4e6642b/edit?viewport_loc=2524%2C3062%2C1918%2C1171%2C0_0&invitationId=inv_56a2d1a5-17ac-45a9-809e-2f07b1152ca1

4. Activity Diagrams

a. Activity diagram (Students)

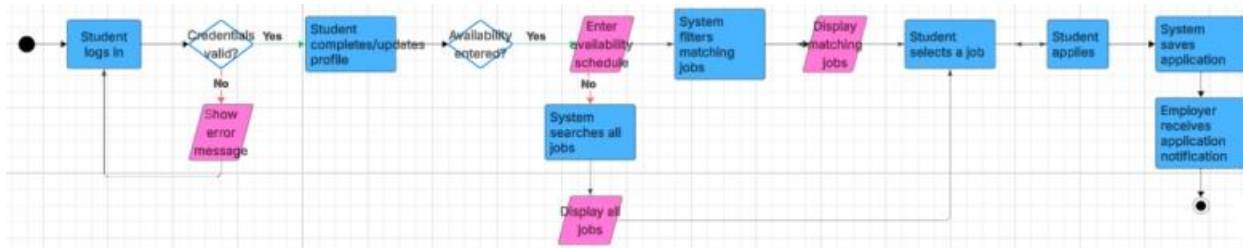


Figure 6: Activity diagram (Students)

b. Activity diagram (Employers)

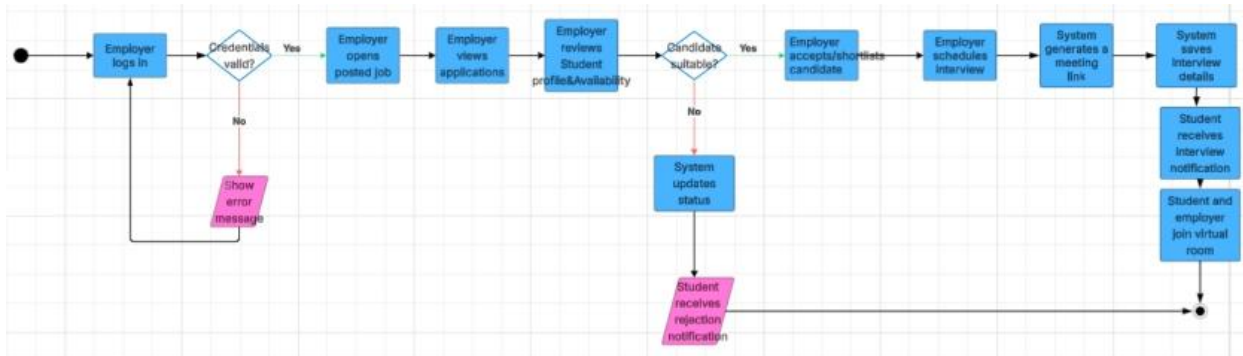


Figure 7: Activity diagram (Employers)

Diagram Link: https://lucid.app/lucidchart/6b772a79-0d96-4c10-9f30-6e47c4e6642b/edit?viewport_loc=2524%2C3062%2C1918%2C1171%2C0_0&invitationId=inv_56a2d1a5-17ac-45a9-809e-2f07b1152ca1

5. State Diagrams

a. Job Application State Diagram

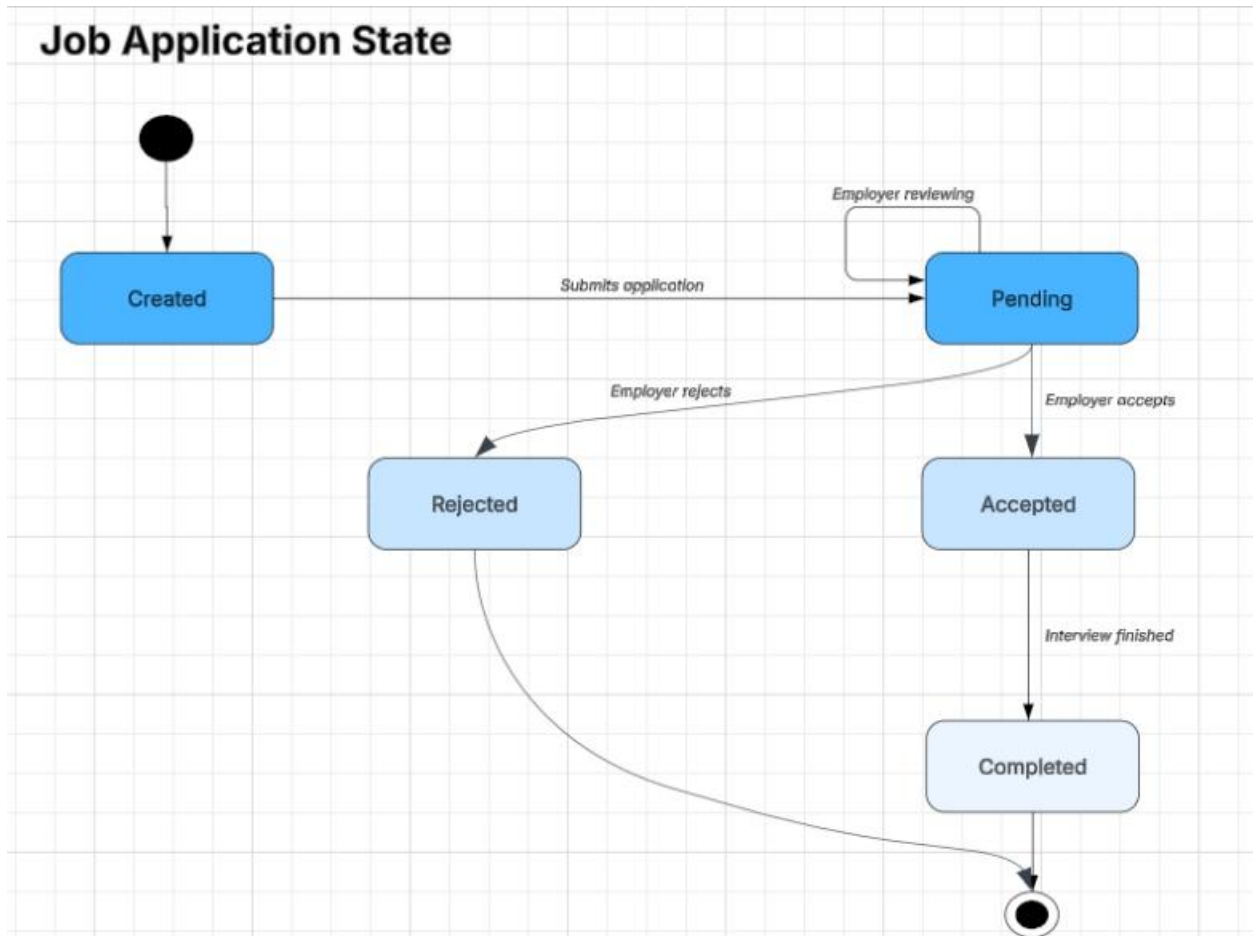


Figure 8: Job Application State Diagram

Diagram Link: https://lucid.app/lucidchart/6b772a79-0d96-4c10-9f30-6e47c4e6642b/edit?viewport_loc=2524%2C3062%2C1918%2C1171%2C0_0&invitationId=inv_56a2d1a5-17ac-45a9-809e-2f07b1152ca1

b. Interview State Diagram

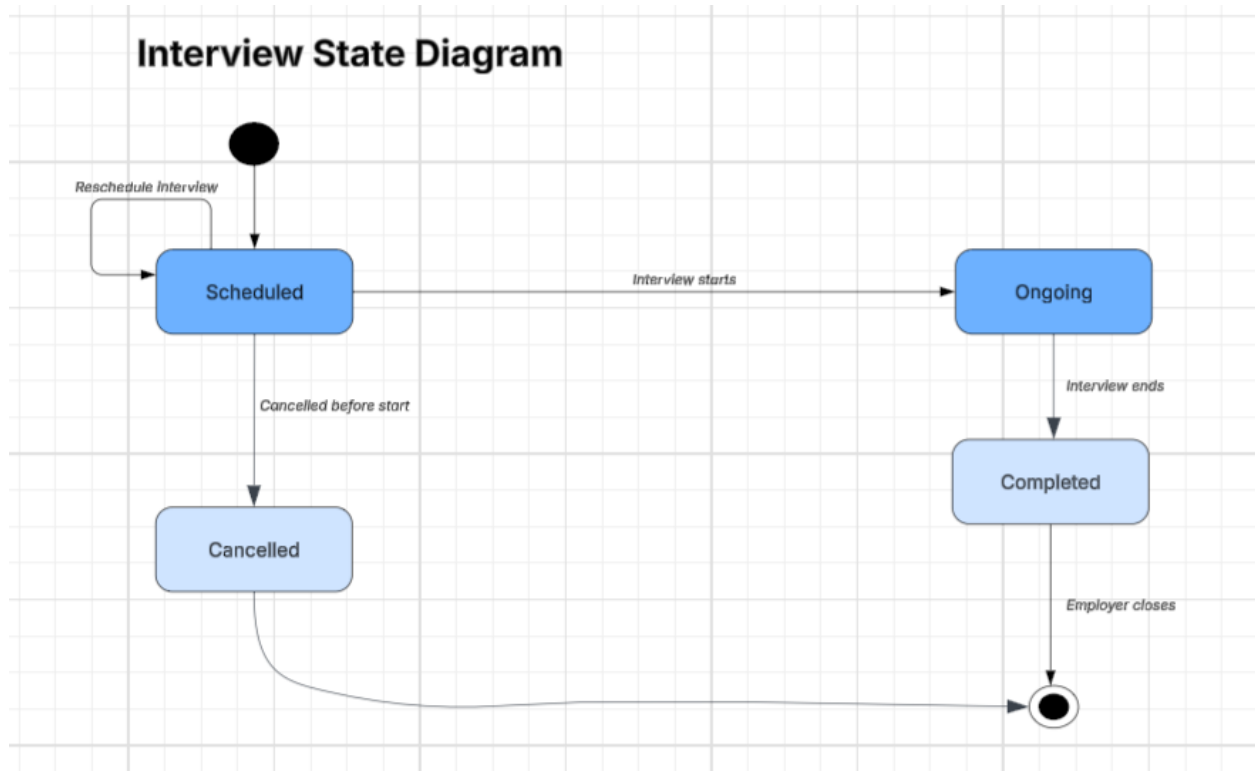


Figure 9: Interview State Diagram

Diagram Link: https://lucid.app/lucidchart/6b772a79-0d96-4c10-9f30-6e47c4e6642b/edit?viewport_loc=2524%2C3062%2C1918%2C1171%2C0_0&invitationId=inv_56a2d1a5-17ac-45a9-809e-2f07b1152ca1

c. *Job Posting State Diagram*

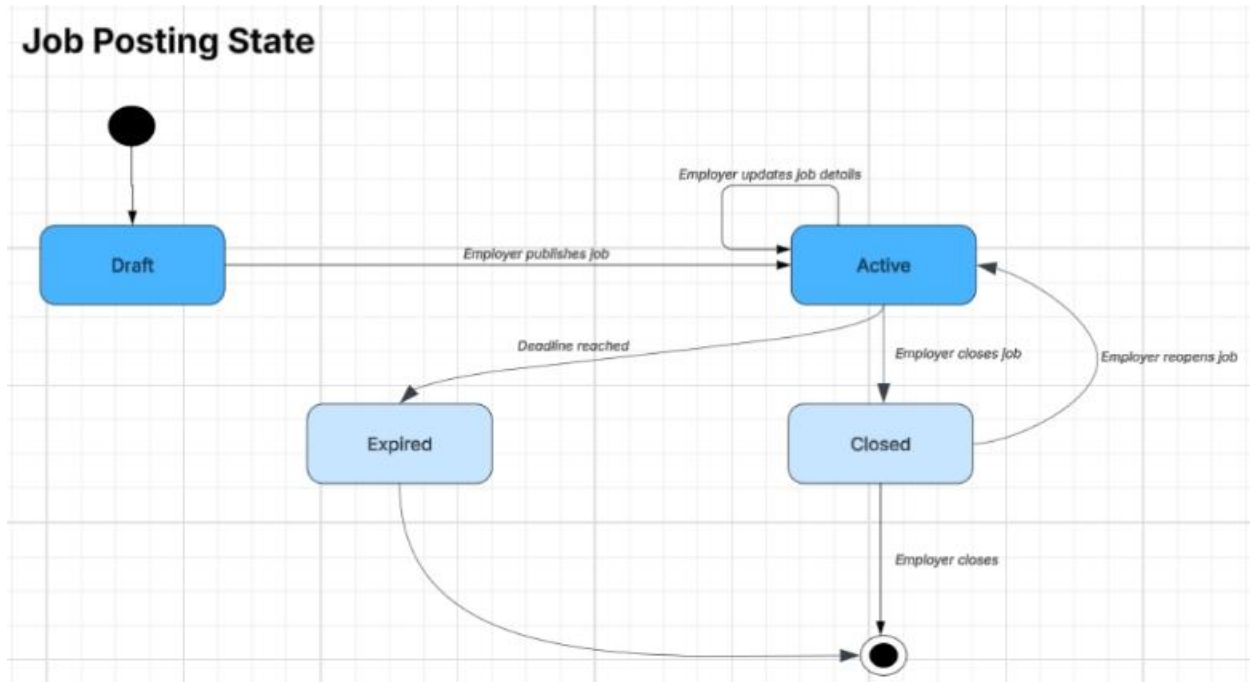


Figure 10: *Job Posting State Diagram*

Diagram Link: https://lucid.app/lucidchart/6b772a79-0d96-4c10-9f30-6e47c4e6642b/edit?viewport_loc=2524%2C3062%2C1918%2C1171%2C0_0&invitationId=inv_56a2d1a5-17ac-45a9-809e-2f07b1152ca1